

E is known.

E' measured

θ fixed experimentally.

vary θ to get θ angle

measure $Q^2 = -4EE' \sin^2 \frac{\theta}{2}$

$$Q^2 = -q^2 > 0$$

measure $\frac{d\sigma}{d\Omega}$ as function of Q^2 / θ

$$\frac{E'}{E} = \frac{1}{1 + \frac{E}{\mu} (1 - \cos \theta)}$$

agreement between data and expected distribution

$$\frac{d\sigma}{d\Omega} \approx \left. \frac{d\sigma}{d\Omega} \right|_{\text{pointlike}} \times |F(Q^2)|^2$$

$$\frac{d\sigma}{d\Omega} \propto \frac{\text{\# events}}{\text{solid angle covered by detector}}$$

a) proton: no structure
no spin.

pointlike

b) no structure $F_1(Q^2) = 0$.
spin with $g=2$ $F_2(Q^2) \neq 0$

c) no structure $F_1 \neq 0$
 $g \neq 2$ anom. mag. moment

Look at data with $F_1 \neq 0$

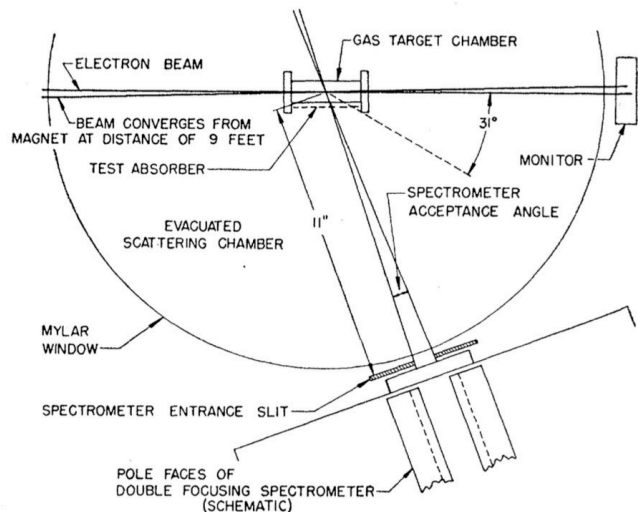
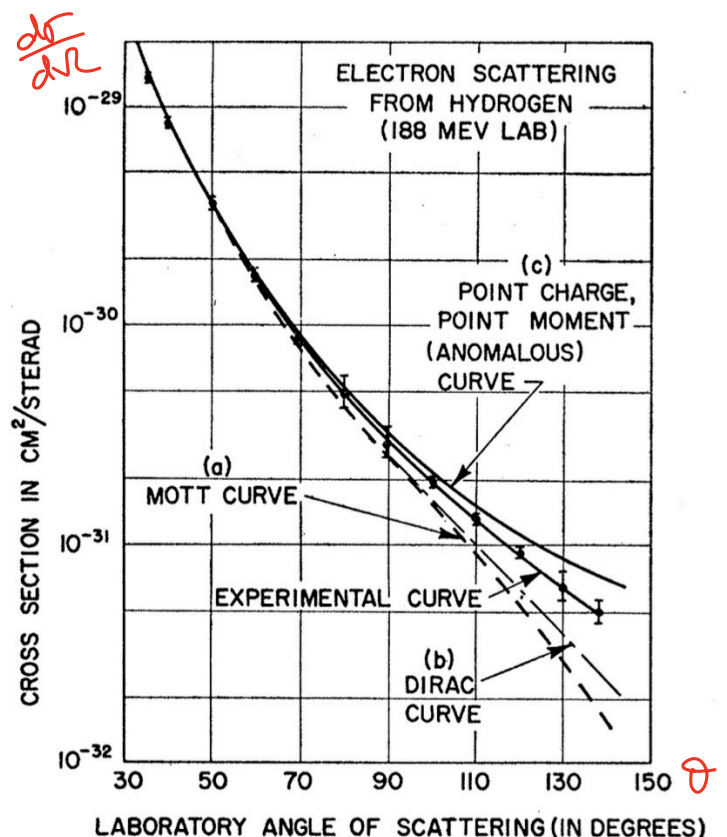


FIG. 2. Arrangement of parts in experiments on electron scattering from a gas target.



$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{struct.}} \approx \left. \frac{d\sigma}{d\Omega} \right|_{\text{point}} |F(Q^2)|^2$$

$$Q^2 = 4EE' \sin^2 \frac{\theta}{2}$$

$$F(Q^2) \neq 0 \quad Q^2 \approx 0 \text{ (small)}$$

$$F_1(Q^2) \approx 1 - \frac{1}{6} q^2 \langle r^2 \rangle$$

$\langle r^2 \rangle$: RMS of spatial distribution

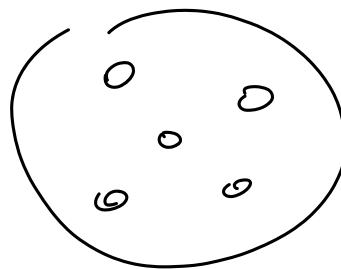
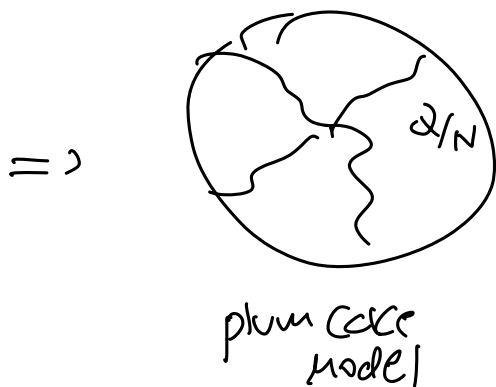
$$\langle r^2 \rangle \sim \int d^3r \rho(r) r^2$$

$$\bar{r} := \sqrt{\langle r^2 \rangle}$$

$$\sqrt{\langle r^2 \rangle} = 0.78 \times 10^{-13} \text{ cm} = 0.78 \text{ fm} > 0$$

From nuclear phys. cs: $r_N = r_0 A^{1/3}$, $r_0 \approx 1.1 \text{ fm}$

$\sqrt{\langle r^2 \rangle} > 0$ (not compatible with $\emptyset$) $\Rightarrow$ proton has structure.



bound state of elementary particles

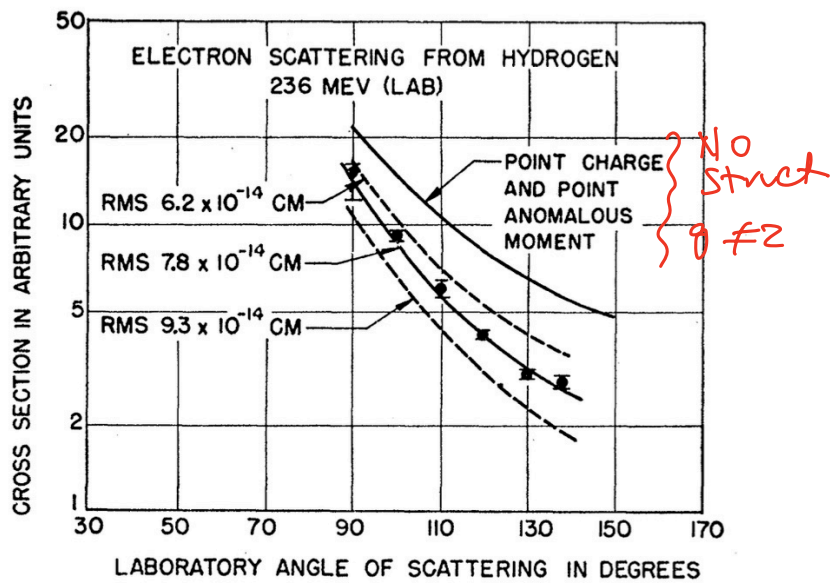
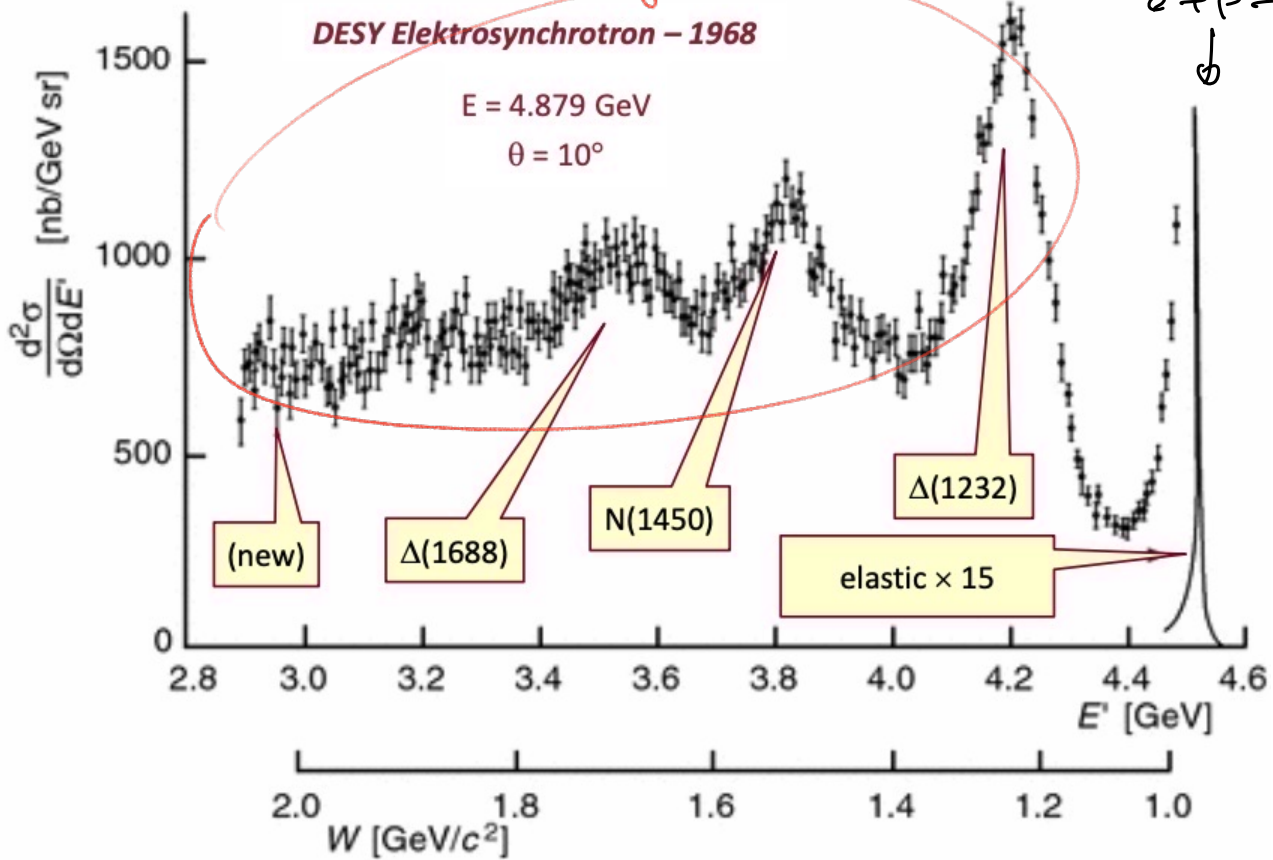


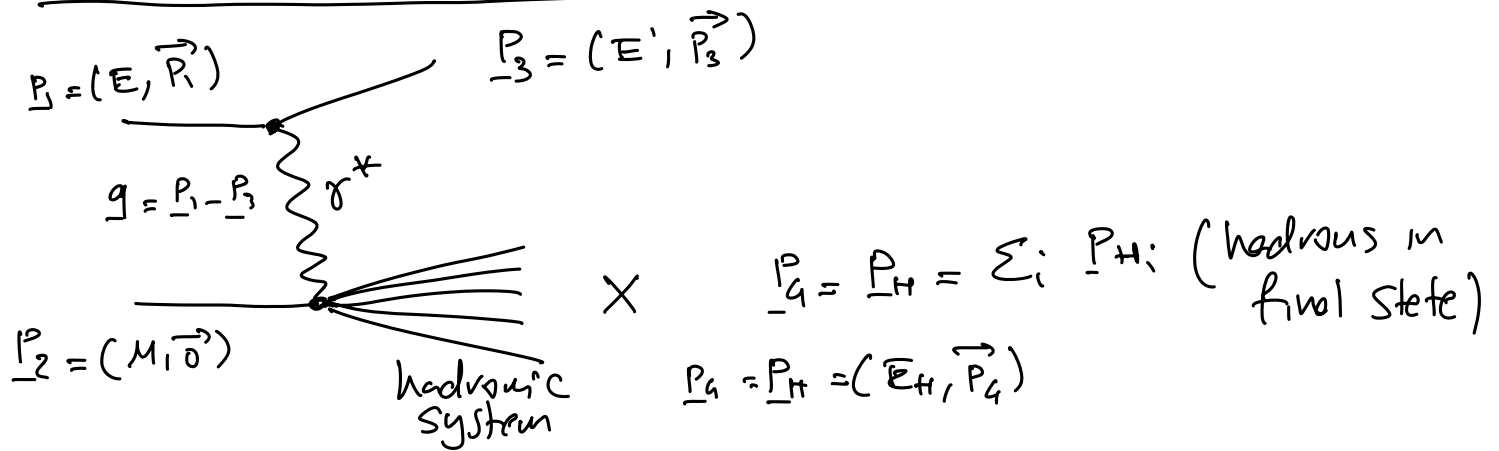
FIG. 6. This figure shows the experimental points at 236 Mev and the attempts to fit the shape of the experimental curve. The best fit lies near $0.78 \times 10^{-13} \text{ cm}$.

$\rho(r)$: charge distribution function.

Inelastic scattering



Inelastic $e^- + p \rightarrow e^- + X$ Scattering



$$q^2 = (\underline{P}_1 - \underline{P}_3)^2 = 2m_e^2 - 4EE' \sin^2 \frac{\theta}{2} \approx -4EE' \sin^2 \frac{\theta}{2}$$

$E \gg m_e$

$Q^2 =: -q^2 \geq 0$

$W^2 =: |\underline{P}_4|^2 = |\underline{P}_H|^2$ invariant mass of hadronic system

$$\underline{P}_4 = \underline{P}_2 + \underline{P}_1 - \underline{P}_3 = \underline{P}_2 + \underline{q}$$

$$(\underline{P}_4)^2 = \underline{P}_2^2 + \underline{q}^2 + 2 \underline{P}_2 \cdot \underline{q} = M^2 - Q^2 + 2M(E - E')$$

$\stackrel{W^2}{=}$

$$W^2 = M^2 - Q^2 + 2M\nu$$

$$Q^2 = -q^2 = 4EE'\sin^2\frac{\theta}{2}$$

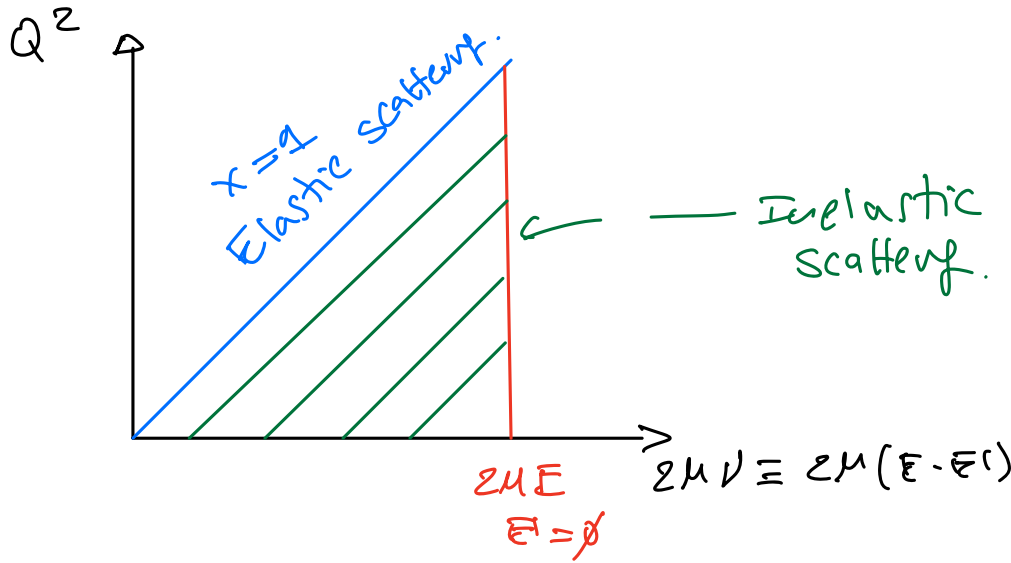
$$\nu = E - E' \geq 0$$

Kinematic description of
Inelastic scattering

Exp. measure: E, E', θ

$$Q^2 = M^2 - W^2 + 2M\nu$$

$$\text{Elastic: } M^2 \equiv W^2$$



Inelastic scattering $M^2 < W^2$

$$Q^2 = \underbrace{M^2 - W^2}_{< 0} + 2M\nu$$

$$x = \frac{Q^2}{2M\nu} = \frac{M^2 - W^2 + 2M\nu}{2M\nu}$$

Björken variable
dimensionless.

$$0 \leq x \leq 1$$

$$y = \frac{\nu}{E} = \frac{E - E'}{E}$$

$0 \leq y \leq 1$ dimensional.

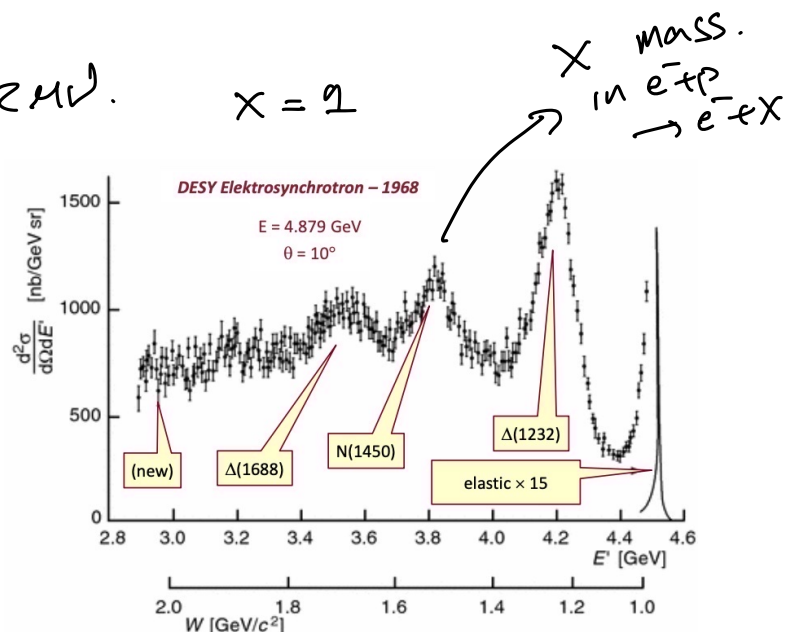
Elastic scattering

$$Q^2 = 2M\nu$$

$$x = 1$$

$$W^2 = M^2 - Q^2 + 2M\nu$$

$$E' \rightarrow 0 \Rightarrow W \uparrow$$



Cross Section for $e^- + p \rightarrow e^- + X$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{inelastic}} = \left. \frac{d\sigma}{d\Omega} \right|_{\text{noH recoil}} \times \left(W_2(Q^2, \nu) + 2 W_1(Q^2, \nu) \tan^2 \frac{\theta}{2} \right)$$

Structure Functions

W_1, W_2 as function of Q^2, ν

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{noH recoil}} \approx \frac{\alpha^2}{94} E^2 \cos^2 \frac{\theta}{2} \frac{E'}{E}$$

Measure $E, E', \theta \rightarrow (Q^2, \nu) \rightarrow (x, y)$

$\Rightarrow$ Measure experimentally W_1, W_2 as function of (Q^2, ν)